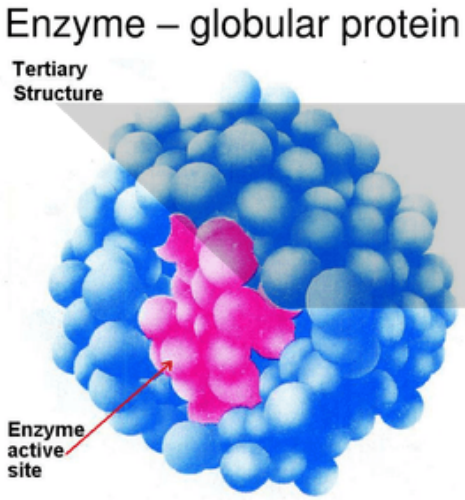
1. Enzymes

1.1. Overview of Enzymes

Enzyme:

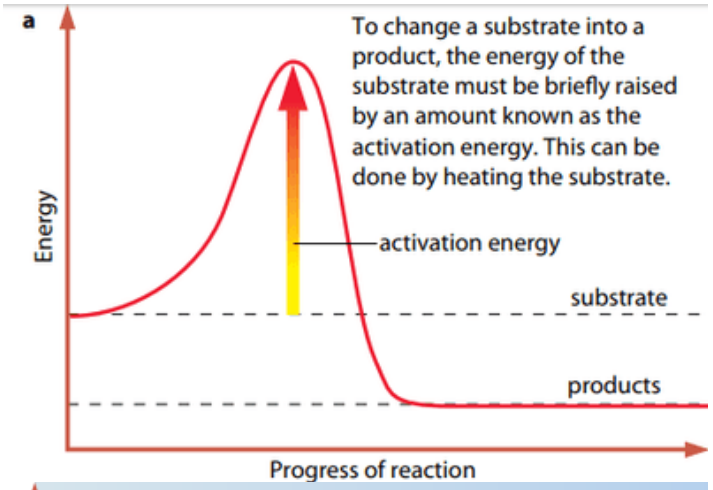
- Biological catalyst that accelerates metabolic reactions
- Remains unchanged at the end of the chemical reaction
- Globular proteins have a roughly spherical shape and are water-soluble

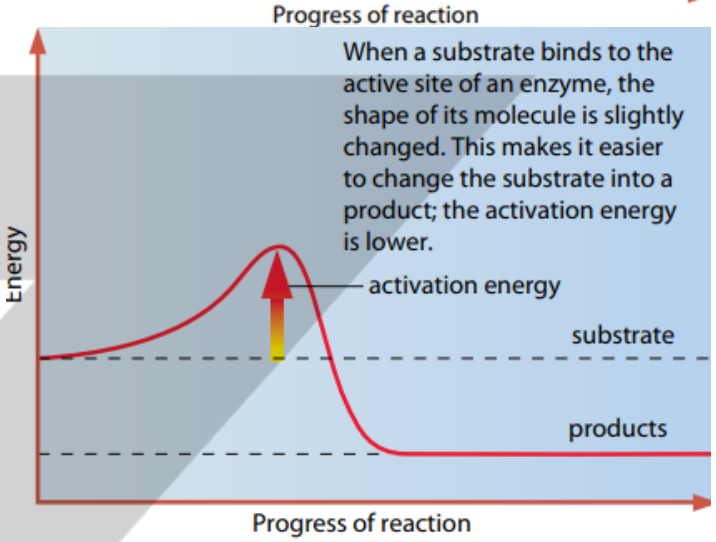


- Enzymes functioning inside a cell are intracellular, but those that are secreted by cells and catalyse reactions outside cells are described as extracellular.

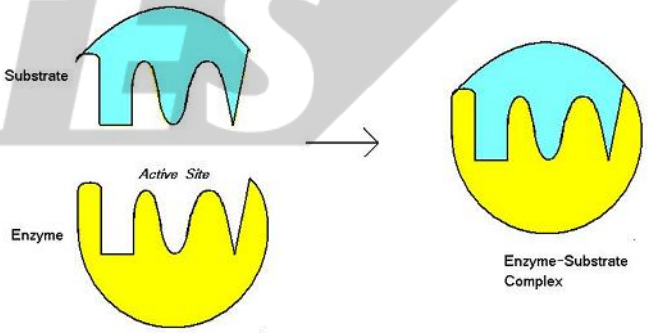
1.2. Mode of Action of Enzymes

- Enzymes have specific active sites that are complementary to the shape of the substrate
- The substrate is held in place at the active site by weak hydrogen and ionic bonds
- The combined structure is called the enzyme-substrate complex
- Activation energy:
  - The energy required in any chemical reaction to break the bonds in reactant molecules so that new bonds are formed to make the product
  - An enzyme lowers the activation energy needed for the reaction
  - However, overall energy released during the reaction is maintained
  - By lowering the *Ea* (activation energy) of a reaction, enzymes increase the rate of the reaction
  - They do this by holding the substrate or substrates in such a way that their molecules can react more easily





- Lock-and-key Theory:
  - The shape of the active site is very precise, and substrates that are not complementary to the shape of the active site cannot bind.
  - The enzyme-substrate complexes formed enable the reaction to take place more easily.



Lock-and-key Model.- The substrate and enzyme active site have complementary shapes

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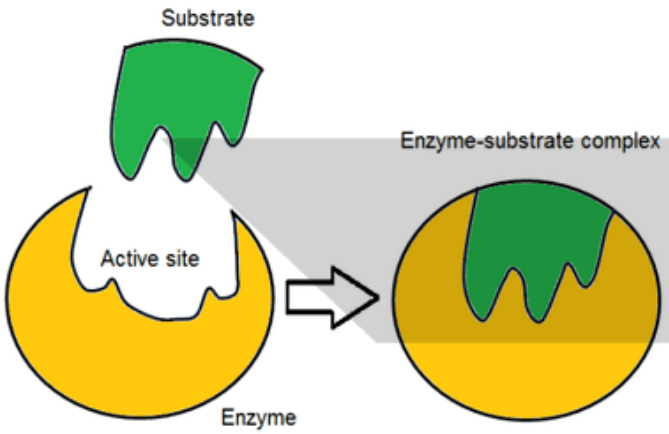
- Induced Fit Theory
  - The enzyme's active site is not initially an exact fit to the substrate molecule.
  - However, the enzyme molecules are more flexible and can change shape slightly as the substrate enters the enzyme.
  - This means the enzyme molecule will undergo conformational changes as the substrate combines with the enzyme's active site, forming the enzyme-substrate complex.

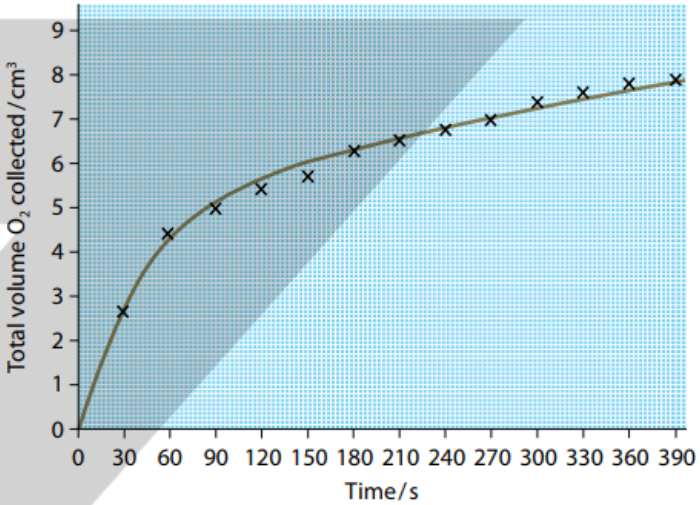


1.3. Investigating the Progress of Enzyme-catalysed Reactions

- The effect that enzymes have on the rate of reactions can be measured in two ways:
  - By measuring the amount of product accumulated over some time
    - Rate of reaction = volume of product produced/ time, e.g. enzyme catalase breaking hydrogen peroxide to H2O + o2 (divide the volume of oxygen produced by the time)
  - By measuring the rate at which the reactants disappear from the reaction mixture
    - E.g., measuring the rate at which starch disappears when the enzyme amylase is added
- Using a colourimeter for reactions that involve colour changes:
  - A colourimeter is an instrument that measures the colour of the solution by measuring the absorption of different wavelengths of light
  - This gives quantitative results which can be plotted on a graph
  - In the amylase/starch experiment, you can measure the intensity of the blue-black colour obtained in the iodine test using a colourimeter
  - The colour acts as a measure of the amount of starch remaining (the lower the colour intensity, the lesser the starch left, and the reaction has progressed considerably, i.e., most of the starch has been converted to maltose)

1.4. The Course of a Reaction

- Initially, there's a large number of substrates, and every enzyme has a substrate in its active site.
- The rate at which the reaction occurs depends only on how many enzymes there are and the speed at which the enzyme can convert the substrate into product, release it, and then bind with another substrate.
- However, over time, there are fewer substrates to bind with enzymes (most get converted to product); the reaction gets slower until it eventually stops.
- The rate of an enzyme-controlled reaction is always the fastest at the beginning.



1.5. Factors that affect Enzyme Action

- Temperature:
  - As the temperature increases, the kinetic energy and the enzyme activity increase as there's more collisions until the optimal temperature is reached (usually 40 degrees C)
  - At optimal temperature, the maximum rate of reaction is achieved
  - If the temperature continues to increase beyond optimal temperature, the rate of the reaction begins to decrease as more kinetic energy breaks the hydrogen bonds in the secondary and tertiary structure of the enzyme
  - This changes the shape of the enzyme and its active site and causes the substrate to no longer fit
  - The enzyme is denatured

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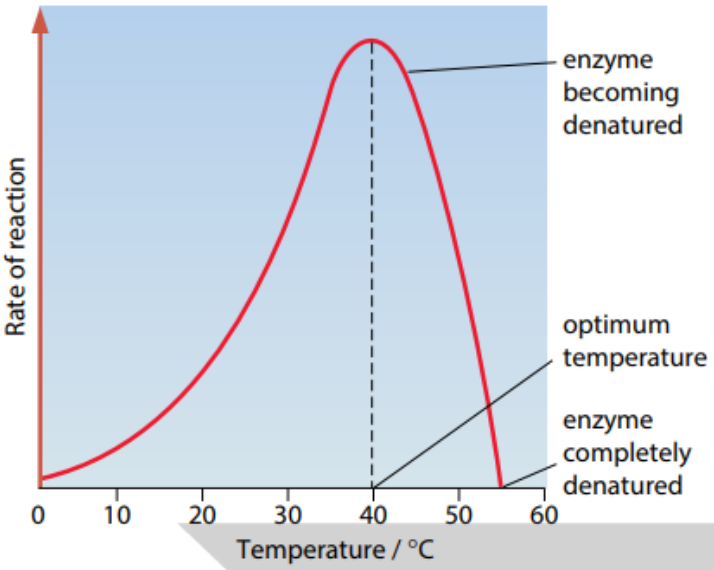
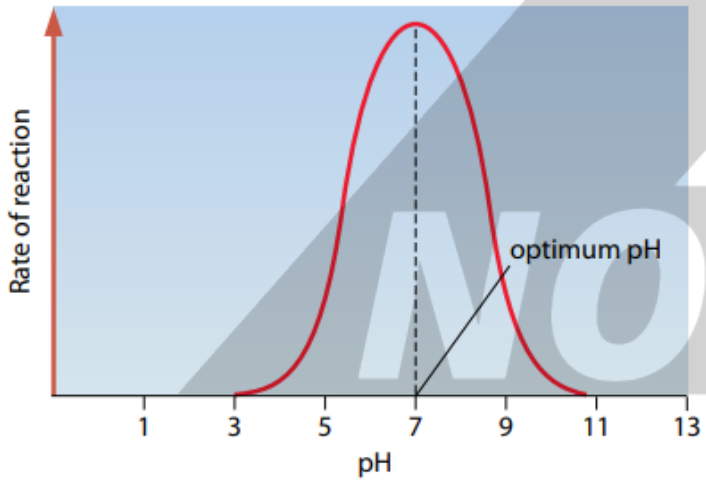
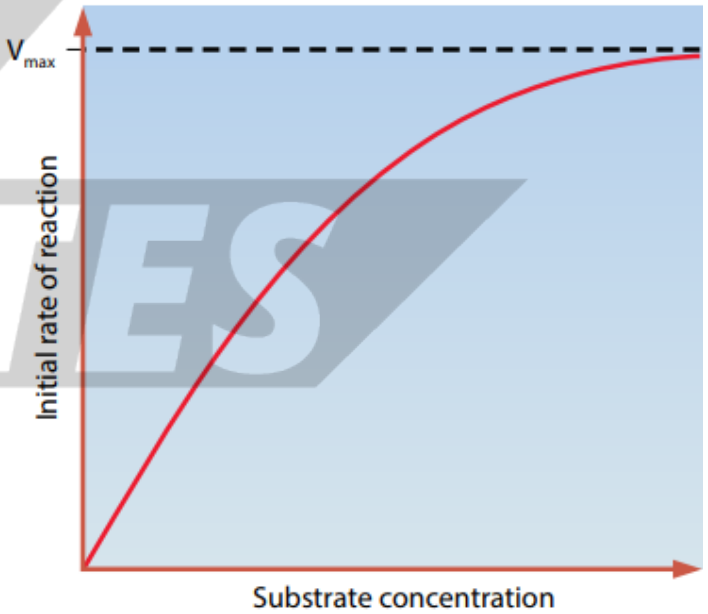


- pH:
  - Any change in the pH value of the medium around the enzyme will cause ionic and hydrogen bonds to be damaged; this will change the 3-D shape of the enzyme and deform the active site
  - The substrate will, therefore not be able to fit into the active site, so the reaction slows down or stops
  - The effects of pH are reversible within certain limits, but if the pH is far from optimal value, the enzyme gets denatured



- Enzyme Concentration:
  - As the concentration of enzymes is increased, there are more available active sites for substrates to fit into
  - More enzyme-substrate complexes are formed, more products are formed, and the rate of reaction is increased
  - The limiting factor is the enzyme concentration
  - Once all substrates have formed enzyme-substrate complexes, a further increase in concentration will not affect the rate of reaction
  - At this point, the limiting factor is the substrate concentration
  - During comparison, look at an initial rate to ensure differences in reaction rate are caused only by differences in enzyme concentration
- Substrate concentration:
  - As the concentration of the substrates increases, there are greater chances of collision with the enzyme
  - More enzyme-substrate complexes are formed, more products are formed, and the rate of reaction is increased
  - The limiting factor is the substrate concentration
  - Once all enzymes are occupied and working at the maximum rate (vmax), a further increase in substrate concentration will not affect the rate of reaction.
  - At this point, the limiting factor is the enzyme concentration.



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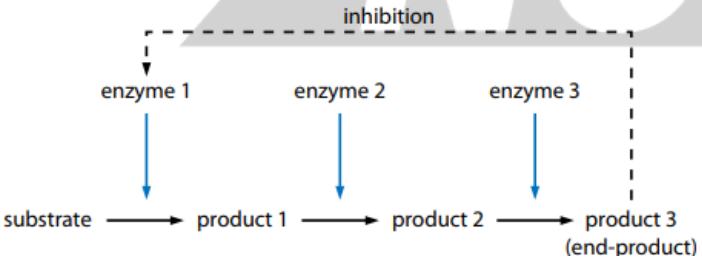
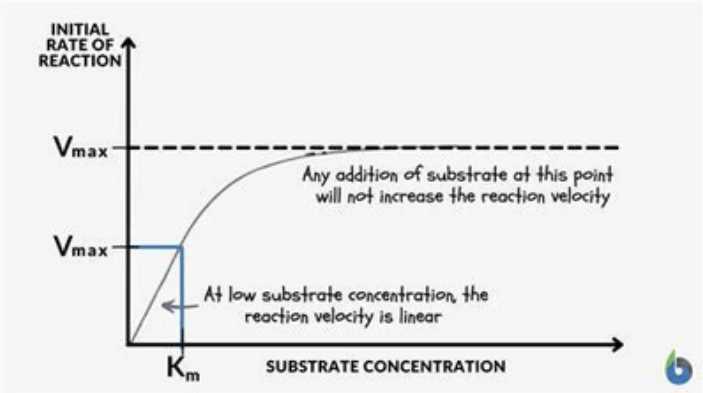
- Inhibitor Concentration:
  - Inhibitors interfere with enzyme activity and reduce the rate of an enzyme-catalysed reaction
  - Therefore, as the concentration of inhibitors increases, the rate of reaction decreases
    - Reversible competitive inhibitor:
      - Has a similar shape to the substrate and fits into the active site
      - This reduces the number of enzyme-substrate complexes formed, and the rate of reaction decreases
      - It is said to be reversible because it can be reversed by increasing the concentration of the substrate
      - Competitive inhibitors increase Km (meaning more substrate is needed to reach half-max speed) but don't change Vmax.
    - The reversible non-competitive inhibitor:
      - Has a different shape from the substrate and fits into a site other than the active site
      - While the non-competitive inhibitor is bound, the tertiary structure of the entire enzyme is distorted, preventing the formation of enzyme-substrate complexes and decreasing the rate of reaction regardless of substrate concentration.
      - Non-competitive inhibitors decrease Vmax (the maximum rate drops) but don't change Km.

1.6. End-product Inhibition

- Used to control metabolic reactions via non-competitive reversible inhibitors
- As the enzyme converts substrate to product, it is slowed down because the end product binds to another part of the enzyme and prevents more substrate binding
- However, the end product can lose its attachment to the enzyme and go on to be used elsewhere, allowing the enzyme to reform into its active state



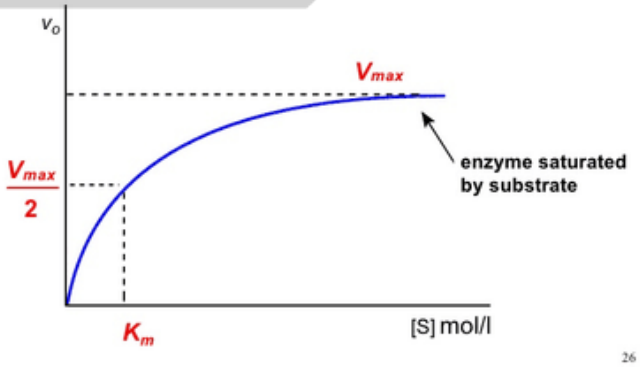
Effect of inhibitors on Vmax and Km



1.7. Maximum Reaction Velocity (Vmax) and Michaelis-Menten Constant (Km)

- Theoretical maximum rate velocity (Vmax):
  - The reaction rate is measured at different substrate concentrations while keeping the enzyme concentration constant
  - As substrate concentration is increased, the reaction rate rises until the reaction reaches its maximum rate
- Michaelis–Menten constant (Km):
  - The substrate concentration that corresponds to half of Vmax is Km
  - Km measures the affinity of the enzyme for the substrate
  - The higher the affinity, the more likely the product will be formed when a substrate molecule enters the active site
  - The higher the affinity of the enzyme for the substrate, the lower the substrate concentration needed for this to happen
  - The higher the affinity, the lower the Km, and the quicker the reaction will proceed to Vmax.

The graph of previous equation is saturation curve



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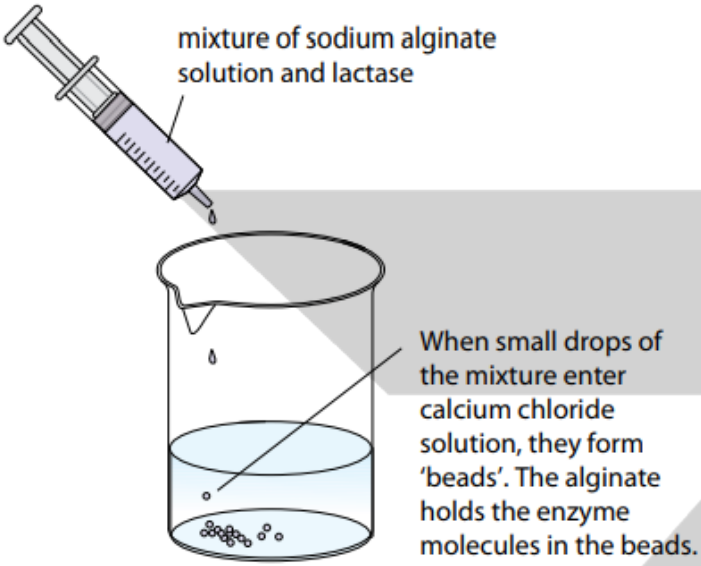
1.8. Enzyme Immobilisation

- The enzyme is mixed with a solution of sodium alginate
- Little droplets of this mixture are then added to a solution of calcium chloride
- The sodium alginate and calcium chloride instantly react to form jelly, which turns each droplet into a little bead
- The jelly bead contains the enzyme



Advantages:

- Can reuse the enzyme easily
- Since the enzyme is not mixed with the solution, it can keep the product enzyme-free, thus preventing contamination
- More tolerant to PH changes as the enzyme molecules are held firmly in shape by the alginate beads, thus not denature easily.
- More tolerant to temperature changes as parts of the molecules embedded in the beads are not fully exposed to temperature or pH changes

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